

Day 2 :- Power Supply Systems Theory : Basics of SMPS and UPS system.
Practical : Identify power cables and connectors.

Tasks 1 :- List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Ans. :-

➤ **Devices Using SMPS (Switched Mode Power Supply) :-**

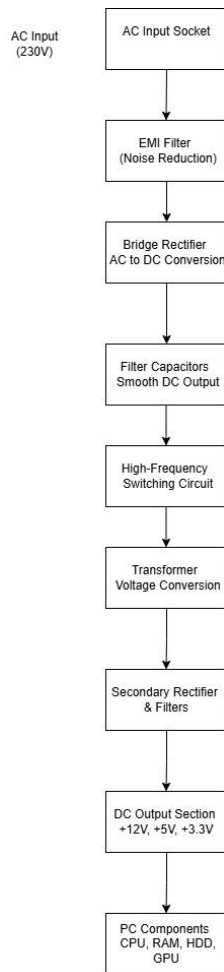
- **Desktop Computer :-** SMPS efficiently converts AC power into the different DC voltages required by PC components.
- **LED Television :-** SMPS provides stable power while reducing energy consumption and heat generation.
- **Laptop Charger :-** SMPS converts mains AC power into regulated DC power for charging the laptop battery.

➤ **Devices Using UPS (Uninterruptible Power Supply) :-**

- **Desktop Computer :-** UPS provides backup power during power cuts, preventing data loss and sudden shutdowns.
- **Wi-Fi Router :-** UPS keeps the internet connection running during power outages for uninterrupted connectivity.

Tasks 2 :- Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.
Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

Ans. :-



Labels :-

- **AC Input** – Receives mains power from the wall socket.
- **EMI Filter** – Removes electrical noise and interference.
- **Bridge Rectifier** – Converts AC power to DC power.
- **Filter Capacitors** – Smooth the DC voltage.
- **Switching Circuit** – Converts DC into high-frequency pulses.
- **Transformer** – Steps voltage up or down and provides isolation.
- **Secondary Rectifier & Filters** – Converts high-frequency AC back to DC.
- **DC Output Section** – Supplies regulated +12V, +5V, and +3.3V to the computer.

Tasks 3 :- Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Ans.:-

ATX 24-Pin Motherboard Connector :-



Devices It Powers :- Main motherboard, chipset, RAM slots, onboard controllers.

Safety Tip :- Turn off and unplug the PSU before connecting or removing it; press the locking clip before disconnecting.

SATA Power Connector :-



Devices It Powers :- SATA HDDs, SATA SSDs, and optical drives.

Safety Tip :- Align the L-shaped connector correctly and never force it into the drive.

Molex 4-Pin Connector :-



Devices It Powers :- Older IDE/PATA hard drives, optical drives, fans, and accessories.

Safety Tip :- Pull the connector by its plastic housing, not by the wires.

PCIe 6/8-Pin GPU Power Connector :-



Devices It Powers :- Graphics cards (GPUs) and other high-power PCIe devices.

Safety Tip :- Pull the connector by its plastic housing, not by the wires.

Tasks 4 :- Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Ans. :-

Steps to Connect the SMPS in a Gaming PC :-

- 1. Turn Off Power**
 - Ensure the SMPS is switched off and unplugged from the wall socket.
- 2. Connect the Motherboard Power**
 - Use the 24-pin ATX power connector from the SMPS.
 - Plug it into the 24-pin ATX socket on the motherboard until it clicks into place.
- 3. Connect CPU Power**
 - Use the 4+4-pin (8-pin) EPS/CPU power connector.
 - Connect it to the CPU power socket near the processor on the motherboard.
- 4. Connect the Graphics Card (GPU)**
 - Install the graphics card in the PCIe x16 slot.
 - Use the PCIe 6-pin, 8-pin, or 6+2-pin power connector from the SMPS.
 - Connect it to the GPU's power socket(s).
- 5. Connect Storage Devices**
 - For SSDs, HDDs, and optical drives, use SATA power connectors from the SMPS.
 - Plug the SATA power connector into each storage device.
 - Also connect a SATA data cable from the storage device to the motherboard.
- 6. Connect Additional Devices (Optional)**
 - Use Molex connectors if required for older fans, RGB controllers, or accessories.
- 7. Check All Connections**
 - Ensure every connector is fully inserted and the locking clips are engaged.

8. Power On the System

- Connect the power cord to the SMPS.
- Turn on the SMPS switch and start the PC to verify that all components are detected.

Connectors Used :-

Component	Connector/Cable Used
Motherboard	24-pin ATX Power Connector
CPU	4+4-pin (8-pin) EPS/CPU Connector
Graphics Card (GPU)	PCIe 6-pin / 8-pin / 6+2-pin Connector
SSD/HDD	SATA Power Connector + SATA Data Cable
Older Fans/Accessories	Molex Connector